

Sri Lanka Institute of Information Technology



Lab Submission Worksheet No 3

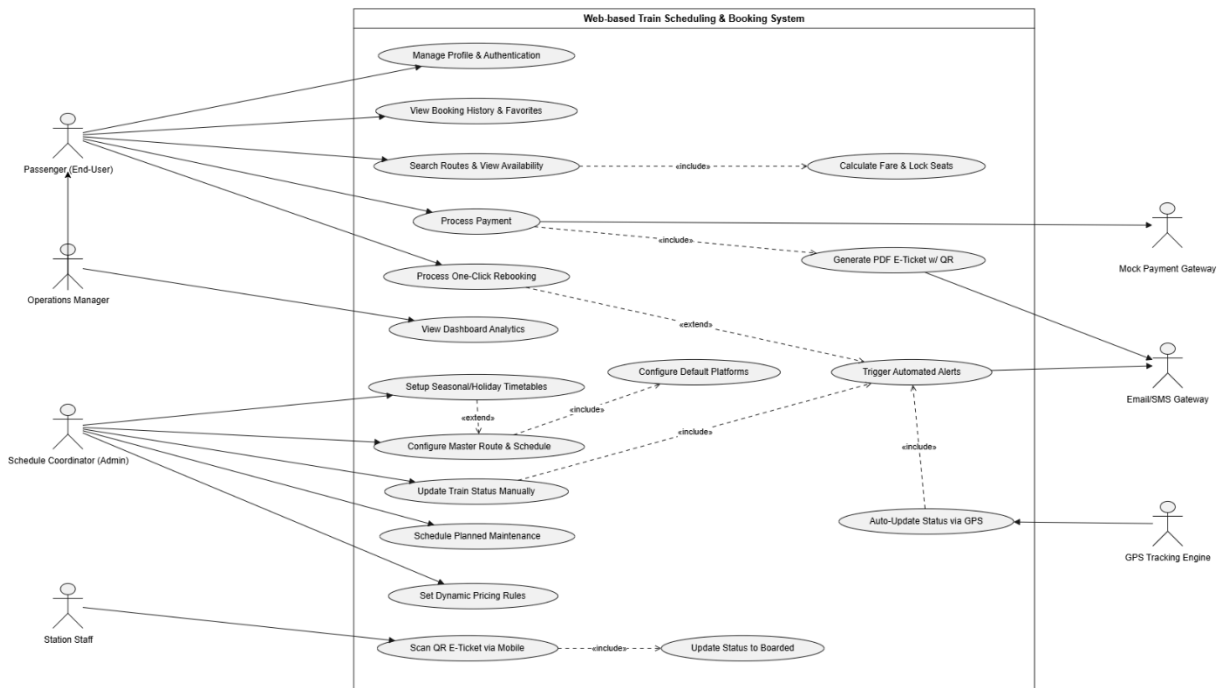
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IT25101520	Dukshanth K.
IT25102327	Balawickrama B.R.D.
IT25103308	Anfas M.S.
IT25100977	Shehara D.M.D.
IT25102925	Sovis W.M.A.V.
IT25100228	Mahanama M.N.S.T.

SE2030 - Software Engineering

B.Sc. (Hons) in Information Technology

Use Case Diagram: Web-based Train Scheduling & Booking System



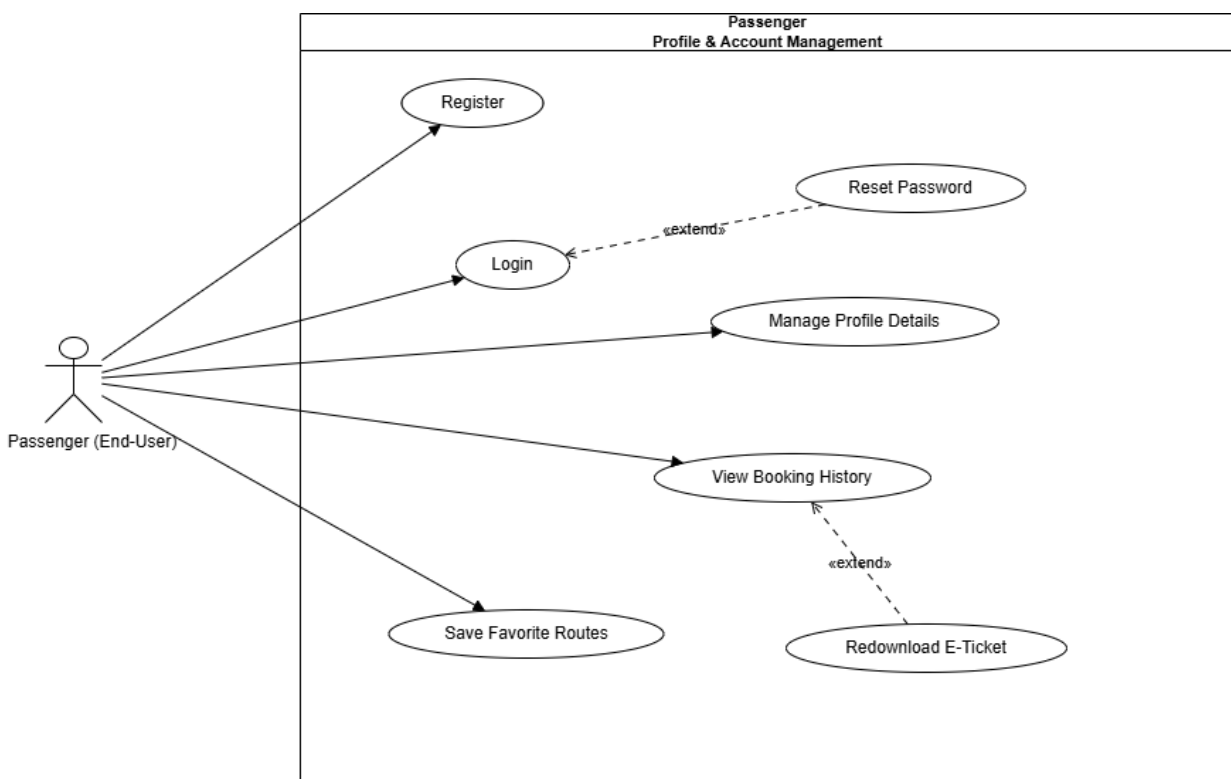
The comprehensive System Use Case Diagram for the Web-based Train Scheduling & Booking System provides a unified, high-level visualization of the entire application's architecture. It consolidates all six major functional modules into a single framework, illustrating the complete interaction network between primary human actors (Passengers, Schedule Coordinators, Operations Managers, and Station Staff), secondary external systems (Payment Gateways, Notification Gateways, and GPS Tracking Engines), and the core system operations. This centralized model demonstrates the system's end-to-end flow—from the initial schedule configuration and seat reservation to real-time status tracking and physical ticket validation at the station.

UC-01: Passenger Profile & Account Management

Dukshanth K. (IT25101520)

A Guest fills out the registration form to create an account. The system verifies that the email is unique, encrypts the password, creates the new user profile, and automatically logs the user in. If the email already exists, the system rejects the registration and prompts the user to log in instead.

Use Case Diagram: Passenger Profile & Account Management



Use Case Scenario: Passenger Profile & Account Management

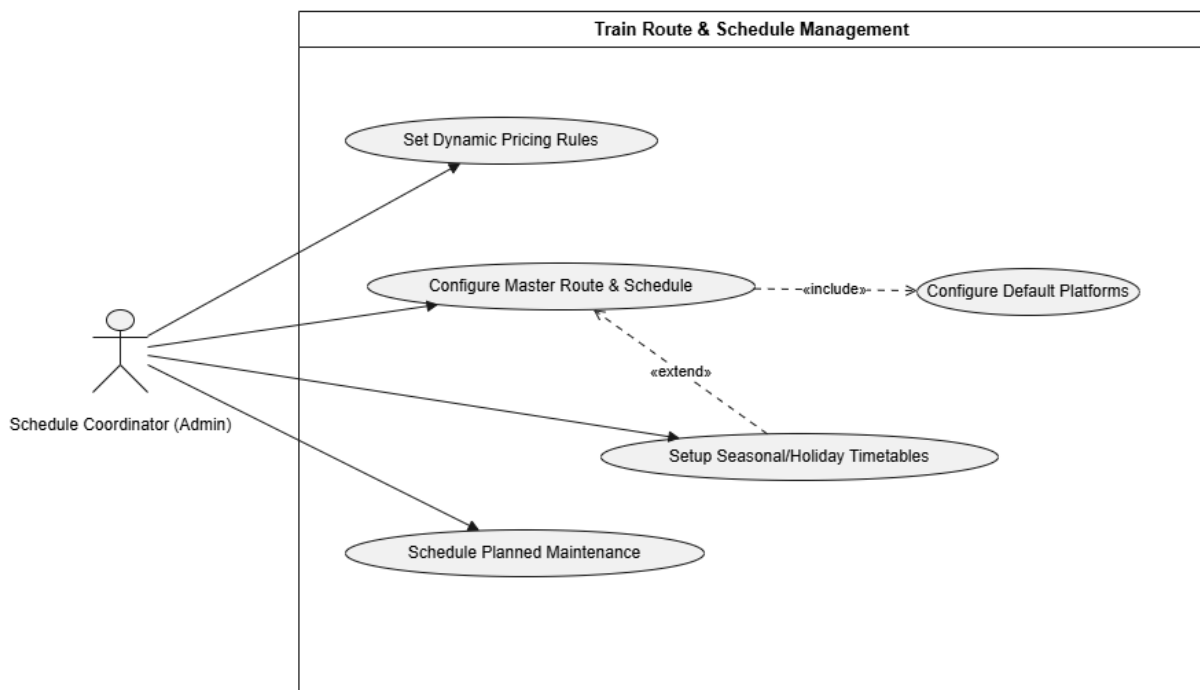
Field	Description
Number	UC-01
Name	Passenger Profile & Account Management
Summary	The passenger manages their authentication, updates personal details, and accesses past booking history.
Priority	4
Preconditions	The passenger has navigated to the system's web portal.
Postconditions	The passenger's profile is created/updated, or their past E-Tickets are retrieved.
Primary Actor(s)	Passenger (End-User)
Secondary Actor(s)	None
Trigger	The passenger chooses to register, log in, or access their profile.
Main Scenario	1. Passenger selects the registration or login option. 2. System prompts for credentials. 3. Passenger provides details and accesses the dashboard. 4. Passenger navigates to booking history or favorite routes. 5. System displays saved contact details, favorite routes, or past E-Tickets.
Extensions	1a. Passenger forgets password; system initiates the password reset flow. 5a. Passenger opts to redownload a past E-Ticket; system provides the PDF.
Open Issues	None

UC-02: Train Route & Schedule Management

Balawickrama B.R.D. (IT25102327)

The Schedule Coordinator enters new train route details (times, stations, and capacity) into the system. The system automatically checks for platform or time conflicts and, if the schedule is clear, saves the new route to the timetable making it instantly available for passengers to book.

Use Case Diagram: Train Route & Schedule Management



Use Case Scenario: Train Route & Schedule Management

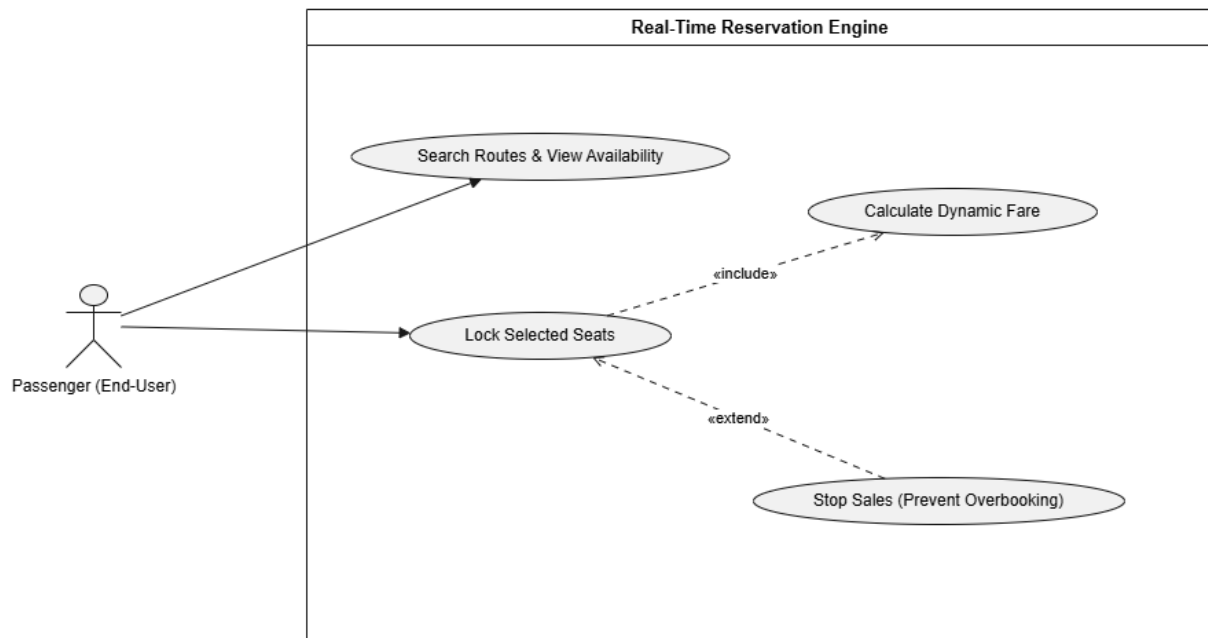
Field	Description
Number	UC-02
Name	Train Route & Schedule Management
Summary	Admin configures master schedules including seasonal variations, pricing rules, platforms, and maintenance blocks.
Priority	5
Preconditions	The Schedule Coordinator is logged into the admin dashboard.
Postconditions	Master timetable, dynamic pricing, and coach capacity are updated globally.
Primary Actor(s)	Schedule Coordinator (Admin)
Secondary Actor(s)	None
Trigger	Admin needs to configure or update the master train network plan.
Main Scenario	1. Admin accesses the Route & Schedule Module. 2. Admin inputs baseline route details, standard stop times, and default platform allocations. 3. Admin configures dynamic pricing rules for the route. 4. Admin sets up Seasonal & Holiday Timetables to automatically override standard schedules on specific dates. 5. Admin defines physical coach capacities and schedules Planned Maintenance Blocks. 6. System saves the configuration and updates the central database.
Extensions	5a. A maintenance block conflicts with existing bookings; system flags the capacity shortage for review. 6a. Admin modifies an active schedule; system temporarily suspends bookings for affected dates.
Open Issues	How far in advance should seasonal timetables be locked to prevent passenger confusion?

UC-03: Real-Time Seat Reservation & Capacity Engine

Anfas M.S. (IT25103308)

A Registered Passenger selects a seat, prompting the system to temporarily lock it for 10 minutes to prevent another user from double-booking. Upon successful checkout, the seat is permanently marked as 'Booked' in the database; however, if the 10-minute timer expires before payment, the system automatically releases the seat lock.

Use Case Diagram: Real-Time Seat Reservation & Capacity Engine



Use Case Scenario: Real-Time Seat Reservation & Capacity Engine

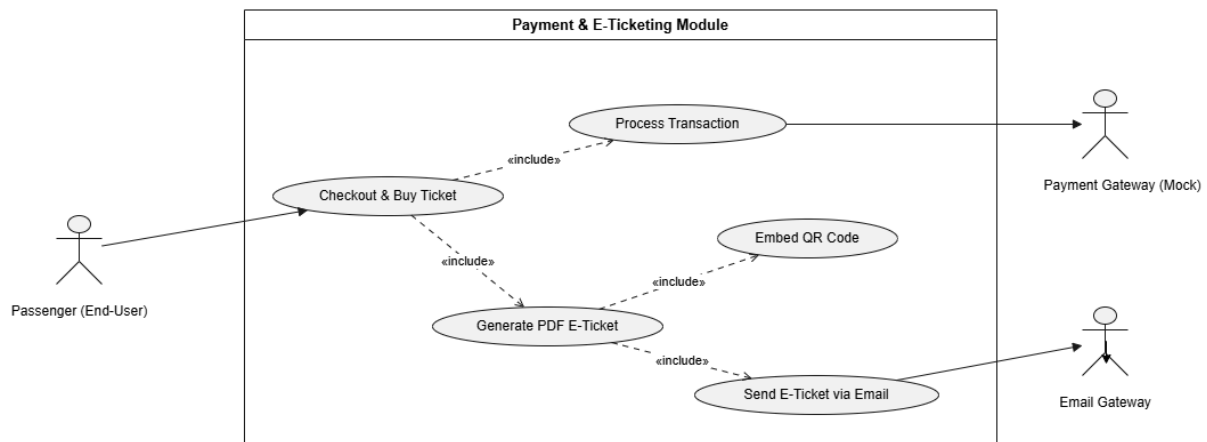
Field	Description
Number	UC-03
Name	Real-Time Seat Reservation & Capacity Engine
Summary	Passenger searches for routes, views availability, and locks seats.
Priority	5
Preconditions	Passenger is logged in; Schedule Coordinator has configured the routes.
Postconditions	Selected seats are locked temporarily for checkout to prevent overbooking.
Primary Actor(s)	Passenger (End-User)
Secondary Actor(s)	None
Trigger	Passenger selects a travel date and route.
Main Scenario	1. System displays available routes and seat capacity per class. 2. Passenger selects desired train, class, and seats. 3. System dynamically calculates the ticket fare. 4. System temporarily locks the selected seats and proceeds to checkout.
Extensions	1a. System notifies user that full capacity is reached and stops sales. 4a. Passenger abandons checkout; system releases locked seats.
Open Issues	1. How long should the temporary seat lock last before timing out?

UC-04: Payment Processing & E-Ticket Generation

Shehara D.M.D (IT25100977)

A Registered Passenger submits their credit card details, which the system processes securely via an external Payment Gateway. If successful, the system generates a PDF ETicket with an embedded QR code and emails it to the passenger; if the payment is declined, the booking is aborted.

Use Case Diagram: Payment Processing & E-Ticket Generation



Use Case Scenario: Payment Processing & E-Ticket Generation

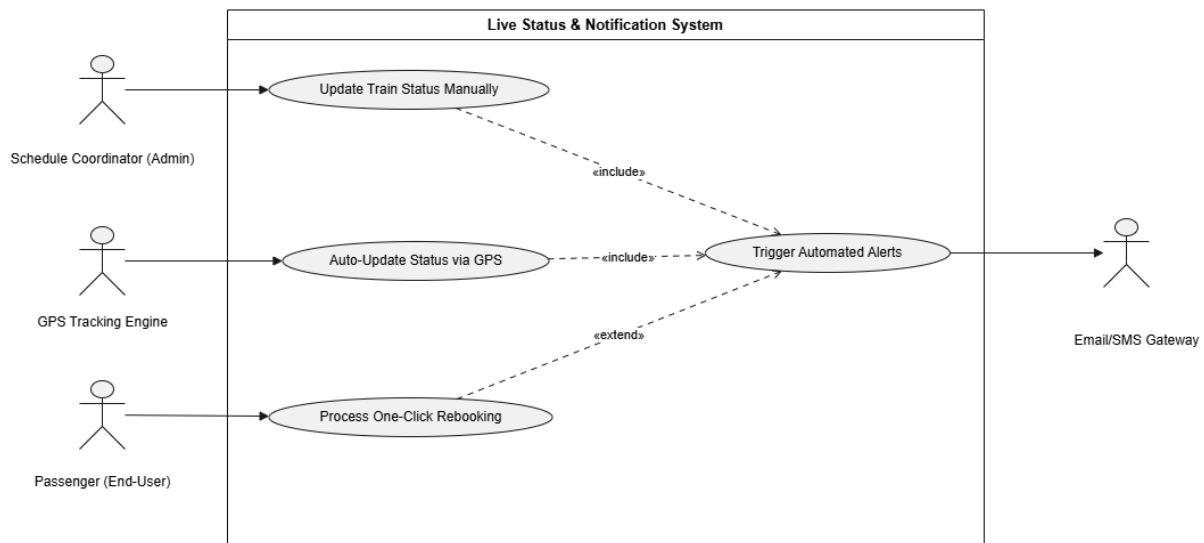
Field	Description
Number	UC-04
Name	Payment Processing & E-Ticket Generation
Summary	The system processes payment securely and generates a scannable PDF E-Ticket.
Priority	5
Preconditions	Passenger has selected and locked their seats in the reservation engine.
Postconditions	Payment is processed, and an E-Ticket is generated and emailed.
Primary Actor(s)	Passenger (End-User)
Secondary Actor(s)	Payment Gateway (Mock), Email Gateway
Trigger	Passenger confirms booking and initiates payment at checkout.
Main Scenario	1. Passenger inputs payment details into the mock payment gateway form. 2. System processes the transaction securely. 3. System generates a downloadable PDF E-Ticket containing booking details and an embedded unique QR code. 4. System automatically sends the E-Ticket via the email gateway to the passenger.
Extensions	2a. Payment transaction fails; system informs the passenger and requests a retry before releasing locked seats.
Open Issues	None

UC-05: Live Status & Automated Notification System

Sovis W.M.A.V. (IT25102925)

The Schedule Coordinator updates a train's operational status to 'Delayed' or 'Cancelled' via the admin dashboard. The system automatically queries the database to identify all passengers with active bookings for that train and sends them an automated warning message via the Notification API.

Use Case Diagram: Live Status & Automated Notification System



Use Case Scenario: Live Status & Automated Notification System

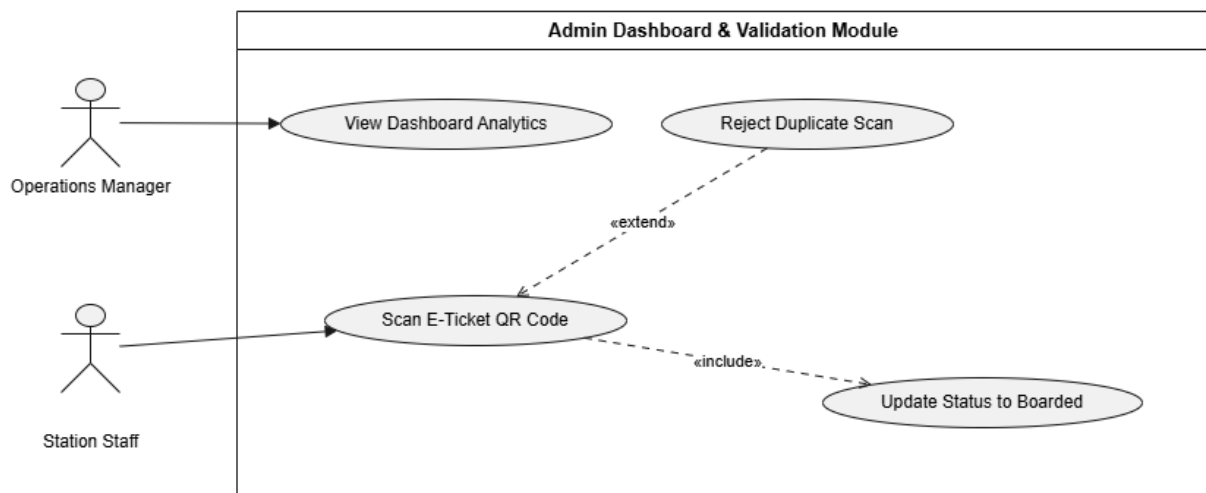
Field	Description
Number	UC-05
Name	Live Status & Automated Notification System
Summary	System tracks live GPS data, updates statuses automatically, alerts passengers, and offers instant rebooking.
Priority	4
Preconditions	Trains are active on the route; passengers have active bookings.
Postconditions	Live status is updated globally, alerts are sent, and rebooking options are provided.
Primary Actor(s)	Schedule Coordinator (Admin), GPS Tracking Engine
Secondary Actor(s)	Passenger (End-User), Email/SMS Gateway
Trigger	Automated GPS tracking detects a delay, or Admin manually cancels a train.
Main Scenario	1. Automated GPS Tracking Simulation continuously monitors train positions and detects a schedule deviation. 2. System auto-updates the live status and station departure boards. 3. System filters the central database for affected passengers. 4. System triggers targeted Email/SMS alerts regarding the delay.
Extensions	1a. Admin manually updates status to "Cancelled". 4a. For cancellations, the system's notification includes a One-Click Rebooking Integration link. 4b. Passenger clicks the rebooking link and is instantly transferred to the next available train without additional payment processing.
Open Issues	What is the maximum acceptable latency for the GPS Tracking Simulation to ensure accurate alerts?

UC-06: Admin Dashboard & Station Ticket Validation

Mahanama M.N.S.T. (IT25100228)

Station Staff scan a passenger's E-Ticket QR code at the physical platform using a mobile device. The system instantly verifies the unique Reservation ID and updates the ticket status to 'Boarded', or triggers a red error screen if the ticket is for the wrong train or has already been used.

Use Case Diagram: Admin Dashboard & Station Ticket Validation



Use Case Scenario: Admin Dashboard & Station Ticket Validation

Field	Description
Number	UC-06
Name	Admin Dashboard & Station Ticket Validation
Summary	Managers track daily metrics while station staff validate passenger E-Tickets using mobile QR scanners.
Priority	5
Preconditions	Station Staff are logged into the mobile scanner page. Passenger presents a QR E-Ticket.
Postconditions	Ticket status is updated to "Boarded" and dashboard metrics are refreshed.
Primary Actor(s)	Station Staff, Operations Manager
Secondary Actor(s)	Passenger (End-User)
Trigger	A passenger presents an E-Ticket for station entry/boarding.
Main Scenario	1. Station Staff scans the passenger's E-Ticket QR code using the mobile scanner page. 2. System verifies the ticket data against the central database. 3. System instantly updates the ticket status to "Boarded" to prevent duplicate entry. 4. System feeds this real-time data into the high-level dashboard analytics for the Operations Manager.
Extensions	2a. Station Staff scans a ticket that is already marked as "Boarded"; system rejects the scan to prevent fraud. 2b. Scanner loses internet connection; validation fails as stable internet is assumed as a system limitation.
Open Issues	None